mode in which its ends vibrate in and out relative to its center. That natural mode, like the ringing of a bell or tuning fork or wine glass, has a well-defined frequency. Just as a bell or tuning fork or wine glass can be made to ring sympathetically by sound waves that match its natural frequency, so the bar can be made to vibrate sympathetically by oscillating tidal forces that match its natural frequency. To use such a bar as a gravitational-wave detector, then, one should adjust its size so its natural frequency will match that of the incoming gravitational waves.

What frequency will that be? In 1959, when Weber embarked on this project, few people believed in black holes (Chapter 6), and the believers understood only very little about a hole's properties. Nobody then imagined that holes could collide and coalesce and eject ripples of spacetime curvature with encoded histories of their collisions. Nor could anyone give much hopeful guidance about other sources of gravitational waves.

So Weber embarked on his effort nearly blind. His sole guide was a crude (but correct) argument that the gravitational waves probably would have frequencies below about 10,000 Hertz (10,000 cycles per second)—that being the orbital frequency of an object which moves at the speed of light (the fastest possible) around the most compact conceivable star: one with size near the critical circumference. So Weber designed the best detectors he could, letting their resonant frequencies fall wherever they might below 10,000 Hertz, and hoped that the Universe would provide waves at his chosen frequencies. He was lucky. The resonant frequencies of his bars were about 1000 Hertz (1000 cycles of oscillation per second), and it turns out that some of the waves from coalescing black holes should oscillate at just such frequencies, as should some of the waves from supernova explosions and from coalescing pairs of neutron stars.

The most challenging aspect of Weber's project was to invent a *sensor* for monitoring his bars' vibrations. Those wave-induced vibrations, he expected, would be tiny: smaller than the diameter of the nucleus of an atom [but he did not know, in the 1960s, how very tiny: just 10^{-21} × (the 2-meter length of his bars) $\simeq 10^{-21}$ meter or one-millionth the diameter of the nucleus of an atom, according to more recent estimates]. To most physicists of the late 1950s and the 1960s, even one-tenth of the diameter of an atomic nucleus looked impossibly difficult to measure. Not so to Weber. He invented a sensor that was up to the task.